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Declining US dental amalgam restorations in US Food and Drug Administration–identified populations: 2017-2023

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ABSTRACT

Background. In light of the Minamata Convention on Mercury and efforts to phase down dental amalgam use, tracking dental amalgam proportions across US Food and Drug Administration (FDA)–identified at-risk populations is of interest to optimize material selection aligned with patient needs.

Methods. A retrospective observational study of Epic's Cosmos electronic health records data set was conducted to calculate the rates of dental amalgam restorations from 2017 through 2023 and stratified using the social vulnerability index (quartile 4 indicates the highest social vulnerability and quartile 1 indicates the lowest) and payer type (Medicare, Medicaid, self-pay, miscellaneous or other). The authors included the following FDA-identified at-risk populations: pregnant people, children younger than 6 years, people with preexisting neurologic conditions, and people with impaired kidney function (n = 1,897,976).

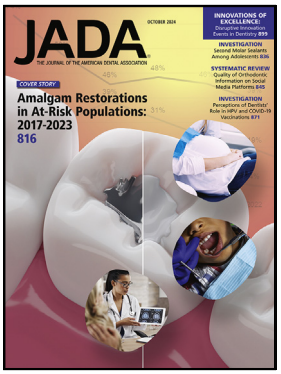
Results. The overall rate of dental amalgam restoration placements in the general population declined from 21.8% in 2017 to 4.1% in 2023. Dental amalgam restoration trends, according to social vulnerability index quartile and payer type, decreased consistently across all 4 evaluated populations. Of all the social vulnerability index quartiles, quartile 4, representing the most socially vulnerable group, had the smallest decrease in dental amalgam placement rates among the FDA-identified populations examined.

Conclusions. The study results showed a decreasing trend in dental amalgam restorations from 2017 through 2023 among FDA-identified populations, consistent with the Minamata Convention on Mercury directive for a phasedown in dental amalgam use. Notwithstanding improvements, lingering disparities persist among the most vulnerable population.

Practical Implications. Even within the groups identified as most vulnerable to harm, more targeted interventions and strategies are required to improve treatment among the most socially vulnerable.

Key Words. Dental amalgam; dental restoration; mercury; disparities; United States.

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Dental amalgam combines elemental (≈ 50%) mercury and a powdered alloy that consists primarily of silver, tin, copper, and other materials. It has been used as a dental restoration material for more than 150 years due to a range of advantages, including safety, ease of use, affordability, effectiveness, and longevity.[1-3] Results of the 2015-2016 National Health and Nutrition Examination Survey showed that 51.5% of teeth with restorations had been restored using dental amalgam.[4] Despite its long history of use, the safety of mercury in dental amalgam is often questioned. In France, alternative restorative materials for pregnant people have been recommended, and in Germany, Finland, Austria, and Canada reducing or phasing out dental amalgam for pregnant people, children younger than 6 years, and those with impaired kidney functions have been proposed.[5] Some nations, including Sweden and Norway, have instituted a ban on dental amalgam use in dentistry.[6]

The Minamata Convention on Mercury is an international treaty focusing on global efforts to protect human health and the environment from the effects of mercury.[7] One of the provisions in the Minamata Convention is setting national objectives aiming at minimizing dental amalgam use.[8] As a signatory to the Minamata Convention, the United States has agreed to phase down the use of dental amalgam, although results of a 2023 systematic review and meta-analysis showed limited evidence to support important differences among direct restorative materials used in practice, including dental amalgam. The 2023 clinical practice guideline from the American Dental Association included dental amalgam among the restorative materials recommended for moderate and advanced caries lesions on vital posterior permanent teeth requiring class I or II restoration.[3,9]

Due to environmental concerns, the US Environmental Protection Agency promulgated a final rule regarding the use of dental amalgam-separating technology to remove and recycle mercury in wastewater discharged from dental offices into municipal sewage treatment plants with a compliance date of July 14, 2020.[10] In 2020, on the basis of what the US Food and Drug Administration (FDA) described as very limited evidence, the FDA updated its recommendations regarding the use of dental amalgam for caries restoration among populations postulated to be more sensitive to the adverse effects of mercury vapor.[11] These populations include people who are pregnant (and their developing fetuses), people planning to become pregnant or who are nursing, children younger than 6 years, people with preexisting neurologic conditions, people with impaired kidney function, and people with a known allergy to mercury or other components in dental amalgam. People in these groups are advised to avoid having caries restored with dental amalgam when possible and to opt for use of alternative materials for dental restoration.

The curriculum in US dental schools has focused increasingly on resin or composite restorations, decreasing emphasis on dental amalgam restoration placement from 2008 through 2018.[12] A similar transition from dental amalgam to resin or composite restorations has been reported to occur in predoctoral pediatric clinics.[13] The results of a retrospective analysis of dental claims data of privately insured people showed an increase in resin or composite restorations and a decrease in the rate of dental amalgam restorations from 2017 through 2019.[14] An analysis of people with a broader range of insurance coverage that included those insured by Medicaid and Medicare as well as those who self-paid for oral health care reported a 73% decrease in the rate of dental amalgam restoration among people with a posterior restoration from 2017 through 2022.[15] However, to our knowledge, no studies on the trends of dental amalgam restoration among FDA-identified populations is available. We aimed to fill this gap by means of examining the changes in dental amalgam and resin or composite materials used for caries restorations in molars and premolars in FDA-identified populations from 2017 through 2023, stratified according to the social vulnerability index (SVI) and payer type.

METHODS

The data used in our retrospective observational study came from Epic (Epic Systems Co) Cosmos electronic health records, a Health Insurance Portability and Accountability Act[16]–defined limited data set. Cosmos is a collaboration of 236 Epic health systems representing more than 227 million patient records as of January 4, 2024.[17] For our analysis, data were drawn from 35,035,169 dental encounters that occurred within the United States from January 1, 2017, through October 26, 2023, and were further limited to the 1,897,976 encounters with codes from *CDT 2023: Current Dental Terminology*[18]: D2140, D2150, D2160, and D2161, which are the codes for dental amalgam restorations of 1, 2, 3, and 4 or more surfaces.[19]

The demographic information available about the population in Cosmos included age group (ie, 0-5 years, 6-14 years, 15-48 years, and $\geq$49 years), rural-urban commuting area classification (ie, urban or rural), and insurance coverage type (ie, Medicare, Medicaid, self-pay, or miscellaneous or other) (Table 1). SVI was used to provide a social context for people, indicating the relative vulnerability of different US Census tracts.[20] This composite index combined 16 factors into the following 4 related themes: socioeconomic status, household composition and disability, racial and ethnic minority status, and housing type and transportation. We assigned each patient an SVI quartile for our analysis on the basis of their ZIP Code of residence. Quartile 4 indicates the highest social vulnerability, and quartile 1 indicates the lowest.

SlicerDicer (Epic Systems Co), a data exploration, analytics, and reporting tool, was used to summarize the dental amalgam and resin or composite restoration encounters using descriptive

ABBREVIATION KEY

FDA: US Food and Drug Administration.
NA: Not applicable.
SVI: Social vulnerability index.

Table 1. Characteristics of the population with dental amalgam and resin or composite restorations from 2017 through 2023.

CHARACTERISTIC	TOTAL,* NO. (%)	WITH DENTAL AMALGAM, NO. (%)	WITH RESIN OR COMPOSITE, NO. (%)
Overall	1,897,976 (100)	182,444 (9.6)	1,715,532 (90.4)
Year			
2017	131,113 (6.9)	28,171 (21.8)	102,942 (79.6)
2018	153,952 (8.1)	28,842 (18.9)	125,110 (82.2)
2019	173,107 (9.1)	32,942 (19.2)	140,165 (81.9)
2020	124,913 (6.6)	17,844 (14.4)	107,069 (86.5)
2021	307,944 (16.2)	25,577 (8.3)	282,367 (92.2)
2022	497,829 (26.2)	28,003 (5.6)	469,826 (94.7)
2023	509,118 (26.8)	21,065 (4.1)	488,053 (96.1)
Age Group, Y			
0-5	43,571 (2.3)	4,899 (11.2)	38,672 (88.8)
6-14	349,360 (18.4)	37,602 (10.8)	311,758 (89.2)
15-48	1,065,210 (56.1)	96,787 (9.1)	968,423 (90.0)
≥49	428,368 (22.6)	43,156 (10.0)	385,212 (90.0)
Social Vulnerability Index Quartile			
1	206,653 (10.9)	9,337 (4.5)	197,316 (95.5)
2	355,000 (18.7)	19,545 (5.5)	335,455 (94.5)
3	448,530 (23.6)	42,722 (9.5)	405,808 (90.5)
4	869,118 (45.8)	110,575 (12.7)	758,543 (87.3)
Rural Urban Commuting Area Classification			
Urban	1,825,091 (96.2)	176,761 (9.7)	1,648,330 (90.3)
Rural	65,742 (3.5)	5,552 (8.4)	60,190 (91.6)
Insurance Coverage Type			
Medicare	9,864 (0.5)	1,255 (12.7)	8609 (87.3)
Medicaid	703,188 (37.0)	75,226 (10.7)	627,962 (89.3)
Self-pay	974,225 (51.3)	14,250 (1.6)	900,408 (98.4)
Miscellaneous or other	110,160 (5.8)	73,817 (43.5)	95,910 (56.5)
Physical Condition			
Pregnancy	21,149† (1.1)	4,004 (19.0)	17,100 (81.0)
Neurologic condition	499,490 (26.3)	63,355 (12.7)	436,135 (87.3)
Impaired kidney function	60,371 (3.2)	8,763 (14.5)	51,608 (85.5)

* Includes dental amalgam and resin or composite restorations. Totals may sum to > 100 because patients can have both types of restorations. † Total pregnancy sum is a rough estimate due to the low sample size for some categories.

statistics, including frequencies and proportions. We used Microsoft Excel to create line charts and bar charts to visualize the data. To avoid patient identification or statistical instability, exact numbers of patients were repressed in table cells with fewer than 11 patients. The analysis further stratified encounters with dental amalgam restorations on the basis of insurance and SVI quartiles for FDA-identified populations. These populations included people who are pregnant (identified as a caries restoration encounter during the same year as a pregnancy), children younger than 6 years, people with preexisting neurologic conditions (*International Classification of Diseases, Tenth Revision, Clinical Modification*[21] codes F01.*, F02.*, F03.*, and R56.*, [* indicates inclusion of any nested codes under that category] or Systematized Medical Nomenclature for Medicine–Clinical Terminology code[22] 118940003), people with impaired kidney function (identified by codes *International Classification of Diseases, Tenth Revision, Clinical Modification* codes N17.*, N18.*, and N19*), and

people with known allergy to mercury or other components of dental amalgam. Given that there is no unique identifier for those with allergies to the components of materials for caries restorations or for people who intend to become pregnant or who are nursing, these populations were not included in our analysis.

RESULTS

The overall general population of people who had dental amalgam and resin or composite restorations in the Cosmos data set during the study period is described in Table 1. Most resided in urban, as opposed to rural, areas (96.2%; n = 1,825,091). Most were in the age group 15 through 48 years (56.1%; n = 1,065,210). Fifty-one percent of the people self-paid for their restorations (n = 974,225), followed by 37% who were enrolled in Medicaid (n = 703,188). Forty-six percent were in the fourth SVI quartile (n = 869,118), the group with the highest social vulnerability. The overall rate of dental amalgam restorations in the general population declined from 21.8% in 2017 to 4.1% in 2023 (Table 1). Conversely, resin or composite restorations increased from 79.6% in 2017 to 96.1% in 2023.

People who were pregnant

Among people who were pregnant and had dental amalgam and resin or composite restorations in the same year, there was a decrease in the proportion of dental amalgam observed across all payer categories (Figure 1). People who self-paid for their restorations had a decrease in the rate of dental amalgam restoration from 74% in 2017 to 24% in 2023. Of dental amalgam restorations in pregnant people, those with costs covered by Medicaid had a continuous decrease from 29% in 2017 to 3% in 2023. The SVI quartiles among pregnant people with restorations indicated a decline in the proportion of dental amalgam restorations among all quartiles from 2017 through 2023 (Table 2). In 2017, quartile 4 had a proportion of dental amalgam restorations at 39%, which decreased over time, reaching 10% by 2023.

Children aged 0 through 5 years

Among children aged 0 through 5 years who received dental amalgam and resin or composite restorations, the rate of dental amalgam restorations for those with Medicaid coverage decreased from 17% of encounters in 2017 to 4% in 2023 (Figure 2). In contrast, for children with miscellaneous or other insurance, the rate increased from 4% to 13%. Self-pay was rare in this group; each year, 10 or fewer dental amalgam restorations for children aged 0 through 5 years were categorized as self-pay.

Children in all 4 SVI quartiles for this age group had a decrease in dental amalgam restorations from 2017 through 2023. Quartile 4 rate of dental amalgam restorations was 12% in 2017, which decreased overall to 8% in 2023 (Table 2).

People with preexisting neurologic disease

For people with preexisting neurologic disease who had dental amalgam and resin or composite restorations, the rate of dental amalgam use declined in all 4 payer categories from 2017 through 2023 (Figure 3). In 2017, people who self-paid for their dental amalgam restorations had a rate of 32%, which decreased to 8% by 2023. The rates among those with Medicaid or Medicare coverage fluctuated during the period examined; in 2017 rates were 24% and 18%, and in 2023 rates were 6% and 7%, respectively.

Considering the dental amalgam restorations according to SVI quartiles, the most vulnerable group (quartile 4) had a rate of 24% in 2017, which decreased to 8% in 2023 (Table 2). Quartiles 1, 2, and 3 had a decrease from 22% to 3.1%, 22.8% to 2.9%, and 28.2% to 5.3%, respectively, from 2017 through 2023.

People with impaired kidney function

Among people with impaired kidney function, the rate of dental amalgam restorations decreased across all 4 payment categories. People who self-paid for their dental restorations exhibited dental amalgam placements of 30% in 2017, which decreased to 11% in 2023 (Figure 4). Rates among people who had claims submitted to Medicare varied from 17% in 2017 to 31% in 2018, followed by a decline to 7% in 2023. Similarly, the rates for miscellaneous or other insurance decreased from

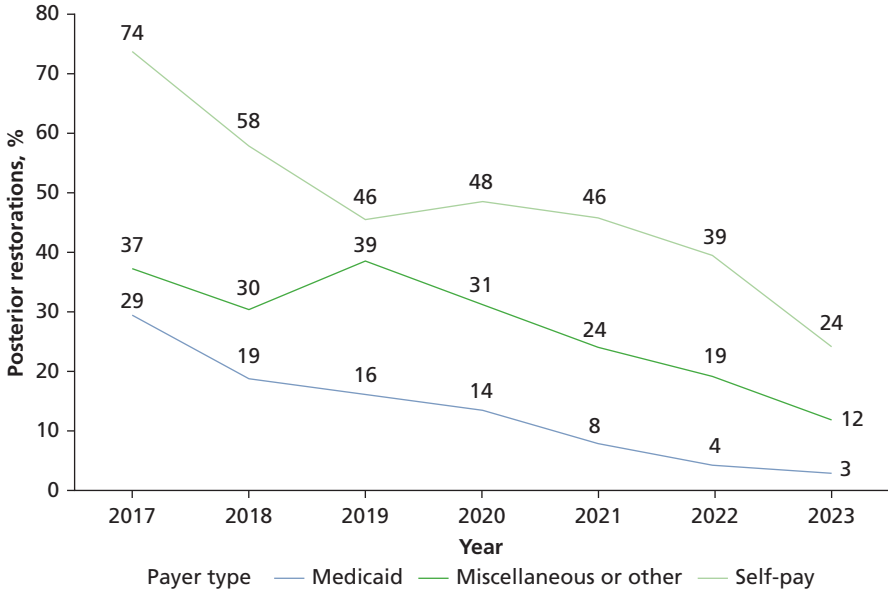


Figure 1. Percentage of posterior restorations with dental amalgam in pregnant people from 2017 through 2023, stratified according to payer type. The number of restorations with dental amalgam for the Medicare group consistently remained at 10 or fewer for all years; hence, it was not included.

Table 2. Dental amalgam restorations stratified according to social vulnerability index quartile for pregnant people, children < 6 years, people with preexisting neurologic conditions, and people with impaired kidney function from 2017 through 2023.

VARIABLE	YEAR, NO. (%)						
Quartile	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**
Pregnant People							
1	27 (25.7)	19 (15.8)	16 (12.1)	≤ 10 (NA*)	11 (6.7)	≤ 10 (NA)	≤ 10 (NA)
2	56 (20.8)	60 (18.8)	70 (18.0)	44 (17.1)	34 (7.3)	26 (4)	18 (2.6)
3	201 (36.3)	154 (26.6)	154 (23.9)	89 (21.5)	160 (19.3)	108 (10.6)	54 (5.2)
4	447 (39.4)	419 (30.3)	683 (36.2)	302 (26.8)	297 (18.1)	338 (14.9)	217 (9.6)
Children < 6 Y							
1	38 (14.2)	24 (8.4)	19 (6.9)	≤10 (NA)	17 (6)	17 (3.5)	18 (4.5)
2	102 (14.2)	100 (12.6)	72 (9.9)	28 (6.7)	47 (6.6)	64 (6.8)	39 (3.6)
3	159 (13.1)	186 (15.0)	174 (14.2)	90 (12)	142 (10.4)	156 (8.4)	126 (7.3)
4	461 (11.8)	551 (13.9)	669 (15.7)	333 (13.0)	433 (11.4)	523 (12.4)	302 (7.9)
People With Neurologic Conditions							
1	642 (22.1)	612 (17.5)	773 (19.0)	396 (12)	588 (7.0)	540 (3.9)	416 (3.1)
2	1,476 (22.8)	1,378 (16.5)	1,407 (15)	789 (11.7)	1,157 (7.8)	972 (4.4)	630 (2.9)
3	2,839 (28.2)	2,701 (22.7)	2,934 (22.3)	1,540 (16.4)	2,173 (11)	2,147 (7.5)	1,431 (5.3)
4	5,678 (23.5)	6,492 (21.3)	6,877 (20.9)	3,574 (16.5)	4,922 (12.8)	5,025 (11)	3,246 (7.6)
People With Impaired Kidney Function							
1	101 (26.2)	109 (21.9)	143 (25.0)	74 (17.2)	96 (9.8)	80 (5.1)	56 (3.9)
2	234 (26.3)	185 (17.3)	205 (17.6)	106 (13.1)	138 (8.5)	127 (5)	80 (3.4)
3	459 (30.9)	392 (24.8)	462 (26.1)	203 (17.3)	300 (12.9)	296 (8.7)	172 (5.6)
4	801 (24.2)	914 (23.1)	932 (21.7)	473 (18.0)	636 (14.5)	582 (11.1)	407 (8.5)

* NA: Not applicable.

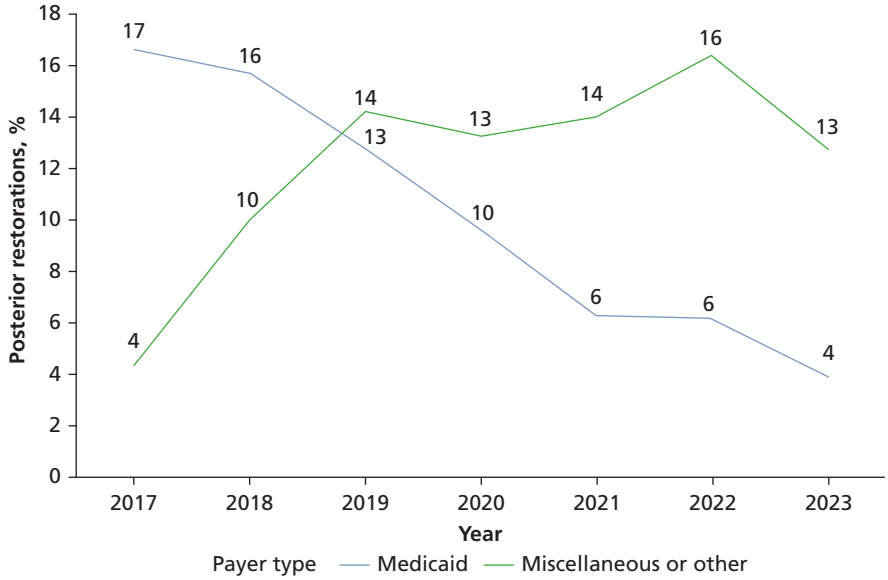


Figure 2. Percentage of posterior restorations with dental amalgam in children aged 0 through 5 years from 2017 through 2023, stratified according to payer type. The number of restorations with dental amalgam for the self-pay group remained at 10 or fewer for all years; hence, it was not included. In addition, as children are covered by Medicaid, Medicare is not represented.

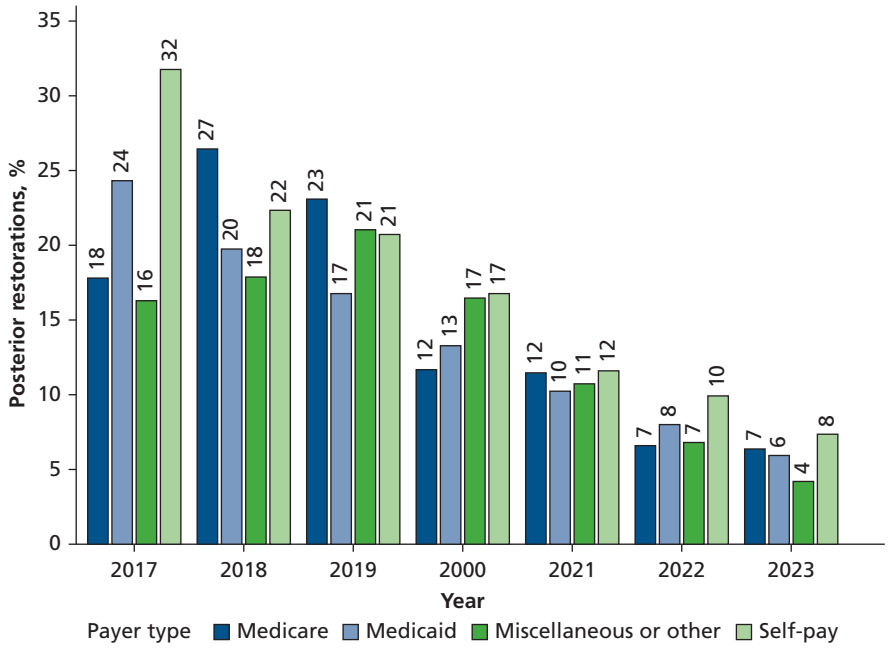


Figure 3. Percentage of posterior restorations with dental amalgam in people with neurologic conditions according to year from 2017 through 2023, stratified according to payer type.

20% in 2017 to 5% in 2023. Rates among those whose restoration claims submitted to Medicaid decreased from 24% of placed restorations containing dental amalgam in 2017 to 8% in 2023.

Looking at the SVI quartiles among the population with impaired kidney function, people in the most vulnerable group (quartile 4) had a proportion of dental amalgam restorations of 24% in 2017 (Table 2). As of 2023, the proportion had reduced further to 9%. The proportion decreased from 26% to 3.9%, 26% to 3.4%, and 31% to 5.6% for quartiles 1, 2, and 3, respectively, from 2017 through 2023.

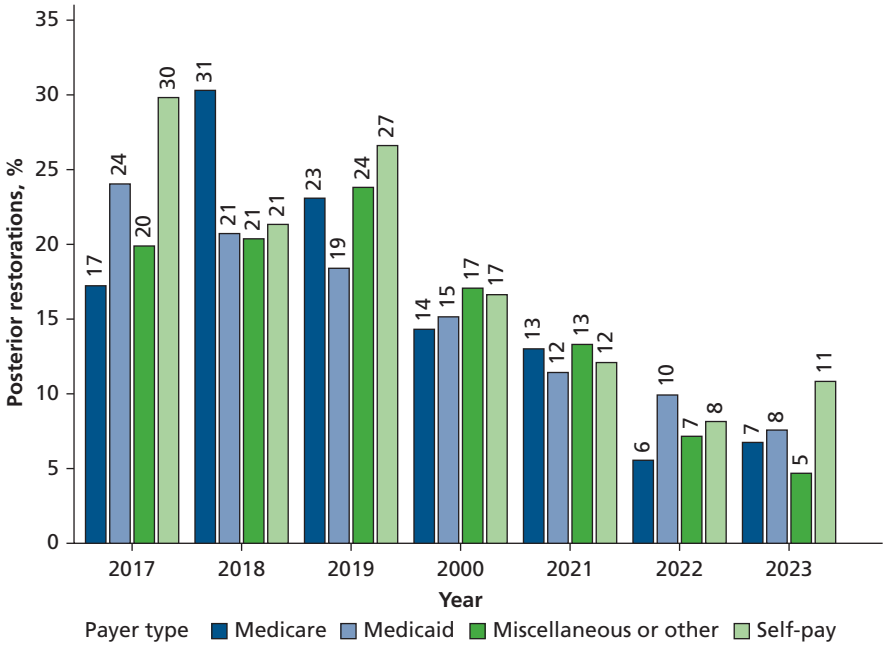


Figure 4. Percentage of posterior restorations with dental amalgam in people with impaired kidney function according to year from 2017 through 2023, stratified according to payer type.

DISCUSSION

Our analyses found the overall proportion of dental amalgam in the general population decreased 81% from 2017 through 2023, consistent with the Minamata Convention on Mercury[7] directive for a phasedown in dental amalgam use. Rather than a change in caries incidence, this reduction appears to be associated more with a change in restorative materials, given the simultaneous increase in the proportion of resin or composite restoration for posterior teeth. This decreased use of dental amalgam coincided with the Environmental Protection Agency's rule on dental amalgam separators in dental offices going into effect and the specific FDA population recommendations.[11,23] The impact of both the rule and the population recommendations is difficult to quantify, although it can be postulated that together they may have contributed to the accelerated decrease in dental amalgam use observed in 2021. The downward trend in dental amalgam restorations is consistent with private insurance claims data from 2017 through 2019 in the United States[14] and international studies[24,25] interpreted as being consistent with the phasedown of dental amalgam use set forth by the Minamata Convention.

We examined dental amalgam restoration patterns within the FDA-identified populations; that is, people who are pregnant, children younger than 6 years, people with preexisting neurologic conditions, and people with impaired kidney function, which were stratified according to SVI and payer type. The stratification according to SVI among these populations indicated an overall decrease in dental amalgam restorations among all 4 quartiles from 2017 through 2023. The smallest decrease occurred among the most socially vulnerable group across all evaluated populations, portraying a nuanced interplay between social vulnerability and dental amalgam restoration. Technological innovation in health care often leads to unequal diffusion, as people of high socioeconomic status are typically the first ones to use and benefit most from early adoption, and those with low socioeconomic status benefit less, creating or exacerbating health inequalities in access to and use of the technologies. These socioeconomic disparities in technology diffusion are observed in various health care areas.[26] This concept might explain our findings that the most vulnerable populations had the smallest declines in dental amalgam use.

The decline from 2017 through 2023 was observed consistently across stratification based on diverse payer types, encompassing Medicare, Medicaid, self-pay, and miscellaneous or other insurance, except for children younger than 6 years. Dental amalgam restorations increased among children younger than 6 years, specifically those covered by miscellaneous or other insurance. This is consistent with the results of a study from Beltrán-Aguilar and colleagues,[27] who reported a higher

mean number of teeth with dental amalgam restorations in the age group 2 through 5 years than nonamalgam restorations (1.6 vs 1.5) from 2015 through 2018. Similarly, the Fair Health National Private Insurance Claims database also found considerably higher rates of dental amalgam restorations among dental patients younger than 6 years than those aged 35 through 55 years.[14] The increased use of dental amalgam as a restorative material among privately insured younger people is likely due to a combination of factors, such as level of caries risk, concerns for recurring caries, insurance coverage and reimbursement policies, parental familiarity, dentist training and office location, and attitudes toward minimally invasive dentistry.[13,28,29] Dentino and colleagues[28] found the odds of having dental amalgam restoration were 14% lower for a 1-surface restoration than a 2-surface restoration, 22% lower for a 1-surface restoration than a 3-surface restoration, and 10% lower for a 2-surface restoration than a 3-surface restoration ($P < .001$). These findings highlight the need for targeted strategies for caries prevention and early lesion detection among younger children, including those unaffiliated with large coverage networks, who may be more reliant on miscellaneous or other insurance. Future research should explore additional factors affecting this specific population.

Among people who were pregnant in the past year or had either impaired kidney function or neurologic conditions, when stratified according to payer type, those who self-paid for their restorations had the highest dental amalgam use (range, 8%-24%) in the year 2023 compared with other payer types. This finding differs from the results of a study of people in Brazil.[30] The low cost of dental amalgam and its durability might be the reasons for higher use of dental amalgam among those who are self-financing their dental restorations.[2,29,31] Targeted interventions to promote the use of alternative materials in self-financing patients who are pregnant or have either impaired kidney function or preexisting neurologic disease may be needed.

A major strength of our study was that it is, to our knowledge, the first study to examine the proportion of dental amalgam restorations among the at-risk populations identified by the FDA. Further stratification according to SVI and payer type contributes to a holistic understanding of socioeconomic factors that may be involved. In addition, the extensive data from Cosmos includes a diverse group of health care institutions representing people in all 50 states, resulting in a comprehensive analysis of changes in dental amalgam use across the US population.

A limitation of our study was its reliance on simple descriptive analysis, precluding drawing causal inferences for the observed changes in the percentage of dental amalgam use over time. In addition, small sample sizes precluded estimation for certain groups, such as children younger than 6 years whose procedures were categorized as self-paid, and introduced volatility to changes over time for other groups (eg, self-paying pregnant people). Another limitation was that we could not incorporate data from November and December of 2023. Although this may slightly underestimate the annual mean, the proportion of dental amalgam to resin or composite restorations should be credible. In addition, we could not distinguish between different classes of restorations (classes I, II, III, IV, V), and it is possible that survival rates may differ depending on the restoration class. There may also be potential assignment errors to groups using ZIP Codes for SVI quartile. Our analysis did not have data to inform an estimate of the use of silver diamine fluoride. Given the prior literature on the increasing trend toward use of silver diamine fluoride, we acknowledge its potential to further decrease the use of dental amalgam, especially in primary teeth, where longevity of restoration is less critical.[32] Lastly, we were unable to assess whether there has been any change in the use of dental amalgam among those with heightened sensitivity to the components of dental amalgam, including silver, tin, copper, and mercury; those who planned to become pregnant in the coming year; or those who were nursing.

CONCLUSIONS

Within the populations studied in this data set, the proportion of dental amalgam and resin or composite restorations using dental amalgam declined from 22% in 2017 to 4% in 2023. An overall decline was observed in all 4 evaluated groups identified by the FDA as at higher risk of developing potential harmful health effects from mercury vapor. Decreases in the dental amalgam restoration placement rate were the smallest among the most socially vulnerable of each of those groups. Future efforts to reduce the use of dental amalgam among children and pregnant people and those with preexisting neurologic conditions or impaired kidney function should concentrate on those who are the most socially vulnerable and include caries prevention strategies in addition to considerations

about restorative materials. Additional studies are required to explore factors associated with informed clinical decision making about oral health and dental material choices. ■

DISCLOSURE

None of the authors reported any disclosures.

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